

Boundary-Spectral Governance — Regime Dossier System (v1.0)

A unified, production-grade framework to diagnose dissipative/contractive dynamics via boundary-spectral structure, time-domain response, and finite-size scaling, with robust confidence scoring and turnkey “Regime Dossiers.”

0) Executive Summary

Goal. Operationalize “zero/boundary” (lawful vs. illegal mass) and classify relaxation regimes with a reproducible, instrumented pipeline:

- **Spectral track:** extremal spectrum, near-zero density, eigenfunction structure.
- **Dynamical track:** decay of illegal mass $\mu(t)$; model selection among exponential, power-law, stretched exponential; multi-exponential decomposition (Prony/ESPRIT).
- **Scaling track:** finite-size behavior of $\lambda_{\min}(N)$ and scaling exponent.
- **Confidence:** weighted agreement across tracks plus disagreement penalty; delocalization metrics incorporated.
- **Deliverable:** a **Regime Dossier** (plots + JSON summary) for each system.

Core regimes.

- **GAP:** $\lambda_{\min}(N) \rightarrow \lambda_* > 0$, exponential tail, localized near-zero eigenfunctions.
 - **CRITICAL:** algebraic scaling $\lambda_{\min}(N) \sim N^{-\beta}$, power-law decay, mixed localization.
 - **CONTINUUM:** $\lambda_{\min}(N) \rightarrow 0$ with delocalized near-zero modes; curved semilog tails.
 - **MULTI-STAGE:** piecewise slopes / multi-band spectra.
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1) System Architecture

Numerical pillars

1. **Spectral analysis (largest N + scaling):**
2. Adaptive thresholds for gap/near-zero region; smallest nonzero eigenvalue $\lambda_{\min}$; density of near-zero modes.
3. **Delocalization diagnostics:** IPR/PR, Shannon entropy, second-moment “localization length.”
4. **Time-domain analysis:**
5. Illegal-mass trajectory $\mu(t) = \|(I - \Pi_{\text{law}})x(t)\|^2$ with $x'(t) = -Ux(t)$ (or the contraction semigroup analogue).
6. Tail model selection via AIC (exp / power / stretched-exp) and **multi-exponential** (Prony + ESPRIT, auto-selected by AIC/BIC).
7. **Finite-size scaling:**

8. Fits for $\lambda_{\min}(N)$ vs. N on log-log; estimate β and λ_{∞} .

9. **Confidence scoring:**

10. Weighted aggregation of **spectral**, **dynamic**, **scaling**, and **delocalization** evidence + disagreement penalty; output overall score + certainty tier.

Sparse-aware & robust. Uses ARPACK for extremal eigenpairs and Krylov $\exp(A)$ actions when available; degrades gracefully to dense paths.

2) Boundary-Spectral Mechanics (MR)

- **Lawful projector** Π_{law} : estimated from near-zero eigenpairs with Gram-Schmidt stabilization; zero/ boundary interprets **conservation** vs. **dissipation sinks**.
 - **Illegal mass** $\mu(t)$: monotone decay under contraction dynamics; primary observable for time-domain validation.
 - **Spectral fork: isolated gap** vs **essential spectrum at 0**; refined to CRITICAL and MULTI-STAGE by scaling and time-domain curvature.
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3) Algorithms & Diagnostics

3.1 Spectral suite

- **Extremal spectrum**: smallest nonzero $\lambda_{\min}$; near-zero density.
- **Delocalization metrics (near-zero subspace):**
- **IPR**: $\sum_i |\psi_i|^4$; **PR**: $1/\text{IPR}$.
- **Shannon entropy**: $-\sum_i p_i \log p_i$, $p_i = |\psi_i|^2$.
- **Localization length**: second-moment width as a 1D proxy.

3.2 Time-domain suite

- **Model fits**: exp / power / stretched-exp via tail regression + AIC.
- **Multi-exponential decomposition**:
- **Prony** (SVD-stabilized Hankel) and **ESPRIT** (shift-invariance, LS/TLS flavor).
- Auto-selection by AIC/BIC; returns rates, amplitudes (and frequencies for ESPRIT).
- **Multi-stage test**: change-of-slope diagnostic on semilog $\mu(t)$.

3.3 Finite-size scaling

- Regress $\log \lambda_{\min}(N)$ on $\log N$; extract β , λ_{∞} , and R^2 .

3.4 Confidence scoring

- Weighted blend (default: spectral .35, dynamic .35, scaling .15, delocalization .15).
 - Disagreement penalty from cross-track indicator mismatch.
 - Output: per-track scores, overall confidence, certainty tier, and agreement score.
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4) Classification Rules (concise)

```
if MULTI_STAGE_indicator: → MULTI-STAGE
elif  $\lambda_\infty > \varepsilon_{\text{gap}}$  and best_model == 'exp': → GAP
elif  $\beta > \beta_{\text{min}}$  and best_model == 'power': → CRITICAL
else: → CONTINUUM
```

ε_{gap} and β_{min} are data-adaptive (via eigenvalue statistics and scaling fit quality) rather than fixed constants.

5) Implementation Artifacts

Core modules produced here

- `boundary_spectral_system.py` — complete, deploy-ready system (delocalization, Prony/ESPRIT, sparse path, dossier generator).
- `mr_boundary_labkit_enhanced.py` — focused lab-kit for spectral/decay analysis and report generation.

Dossier examples (generated)

- **GAP (Dirichlet FD):** `.../regime_dossiers/GAP_fd_dirichlet/`
- **CRITICAL (power-law):** `.../regime_dossiers/CRITICAL_powerlaw/`
- **CONTINUUM (near-zero band):** `.../regime_dossiers/CONTINUUM_band/`

Each dossier contains: `summary.json`, `lambda_vs_N.png`, `mu_semi-log.png`, `mu_log-log.png`, `confidence.png`, `ipr_hist.png`.

6) API Sketches & Quickstart

Single dossier

```
import boundary_spectral_system as bss

res = bss.generate_regime_dossier(
    case_name="YOUR_SYSTEM",
    params=dict(method="fd", bc="dirichlet"), # or custom method wiring
    N_values=(64, 128, 256),
    T=200,
    out_dir="./your_dossiers",
    seed=42,
    use_sparse=True
```

```
)
print(res["summary_json"]) # path to JSON
```

Direct analysis (for orchestration)

```
res = bss.analyze_system_enhanced(
    case_name="YOUR_SYSTEM",
    params={...},
    N_values=(64, 128, 256, 512),
    T=200,
    seed=123,
    use_sparse=True
)
```

Custom generator wiring

- Introduce a new `method="your_method"` branch in `construct_truncation[_sparse]`, or
- Wrap your builder so it returns a (sparse) matrix for each N , then pass `params` selecting it.

7) Validation Protocol (ready-to-run)

1. **Build truncations** at several N with explicit boundary conditions.
2. **Estimate Π_{law}** from near-zero eigenpairs (stability checked; Gram-Schmidt).
3. **Evolve illegal mass** from an orthogonal state; extract $\mu(t)$ tail.
4. **Fit models** (AIC) + **multi-exp** (Prony/ESPRIT); flag multi-stage.
5. **Compute delocalization** metrics on near-zero subspace.
6. **Finite-size scaling** of $\lambda_{\min}(N)$; get β , λ_{∞} , R^2 .
7. **Classify** regime; compute confidence + certainty.
8. **Emit dossier** (plots + JSON) and a run manifest for provenance.

8) Numerical & Practical Notes

- **Stability:** all regressions restricted to tail windows; eigen-subspace orthonormalized.
- **Adaptivity:** thresholds drawn from eigenvalue statistics; time steps set by simple spectral scales.
- **Sparse path:** ARPACK for extremal pairs; Krylov `expm` action for large N ; dense fallback retained.
- **Throughput:** parallelization slots neatly over systems; plotting isolated per figure (no subplots; no style overrides).

9) Roadmap (optional extensions)

- Add **matrix-pencil** and **ESPRIT-TLS** variants with automatic conditioning checks.

- Extend delocalization to **multidimensional** geometries (PR by block-coarse-graining).
- Synthetic-data **benchmark suite** and tiny **ML classifier** over summary features.
- Reproducible **YAML configs + batch runner** for cloud pipelines.

10) Internal Project Notes (for context)

- *Becoming-two vs Being-two — Twoness as Functorial Forgetting.*
- *Where Do Negatives Come From? — Operational Closure, Completion, And Baselines.*
- *Modes of Infinity as Modes of Being — Formalization Blueprint.*
- *Ofa(ii) as Initiality: Free-Monoid Iterators and Recovery of the NNO.*
- *Mr Boundary Theory — Zero as Boundary Operator (roles, μ , and Spectral Tests).*
(Primary conceptual seed for this toolkit.)
- *QPF Methods Protocol (gallows-proof, v1).*
- *Entangled Aggregation: Non-associativity as a Closure Defect under Coarse-grained Composition.*
- *CCN v2 Minimal Axioms + Representation Theorem (finite case).*
- *Modal Numerosity as Algorithmic Inference — camera-ready package.*
- *Aspectual Counting Framework.*

(These PDFs document the metaphysical grounding: boundary, zero, lawful/illegal mass, and counting/initiality perspectives.)

11) External References (selected)

Semigroups, generators, and gaps

- Engel & Nagel, *One-Parameter Semigroups for Linear Evolution Equations*, Springer, 2000.
- Pazy, *Semigroups of Linear Operators and Applications to Partial Differential Equations*, Springer, 1983.
- Lumer & Phillips, "Dissipative operators in a Banach space," *Pacific J. Math.*, 11 (1961).
- Kato, *Perturbation Theory for Linear Operators*, Springer, 1995.
- Lindblad, "On the generators of quantum dynamical semigroups," *Comm. Math. Phys.*, 48 (1976).

Numerical linear algebra & sparse methods

- Lehoucq, Sorensen, Yang, *ARPACK Users' Guide*, SIAM, 1998.
- Al-Mohy & Higham, "Computing the Action of the Matrix Exponential," *SIAM J. Sci. Comput.*, 33(2), 2011.
- Higham, *Functions of Matrices*, SIAM, 2008.

Signal models / multi-exponential estimation

- Prony, "Essai expérimental et analytique," 1795 (classical method).
- Roy & Kailath, "ESPRIT—Estimation of Signal Parameters via Rotational Invariance Techniques," *IEEE Trans. ASSP*, 37(7), 1989.
- Hua & Sarkar, "Matrix Pencil Method," *IEEE Trans. ASSP*, 38(5), 1990.

Localization & participation measures

- Evers & Mirlin, “Anderson transitions,” *Rev. Mod. Phys.*, 80, 1355 (2008).
(Comprehensive review including IPR/PR across dimensions.)
- Abrahams (ed.), *50 Years of Anderson Localization*, World Scientific, 2010.

Scaling & criticality

- Cardy, *Scaling and Renormalization in Statistical Physics*, Cambridge, 1996.

(The above references cover the operator-semigroup backbone, near-zero spectral theory, numerical methods for large systems, and the signal-processing estimators used in multi-exponential decay analysis.)

12) How to Use This Document

- Treat this page as the spec + field manual for running classifications and producing Regime Dossiers.
- Fork the `boundary_spectral_system.py` entry-point and wire in your **real** $\mathcal{U}_N$ constructor (PDE, Lindbladian, Markov chain, etc.).
- Use the dossier artifacts to communicate regime decisions with reproducible evidence.

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